Grade it yourself first

Two answers to a real Calculus I exam question. Based on real answers from my class, rewritten so no student can be identified.

Question (5 points). Evaluate $\lim_{x \to -3} (x^2 + x - 6) / (x^2 + 3x)$, stating the form that substitution gives. Show all work.

1. Substitutes $x = -3$ correctly and shows the result is $0/0$
2. Names the form: the indeterminate form $0/0$
3. Factors correctly: $(x + 3)(x - 2)$ over $x(x + 3)$
4. Cancels $(x + 3)$ and keeps limit notation, reaching $\lim (x - 2)/x$
5. Correct final answer: $5/3$

Each rubric line is worth 1 point.

Answer B

$\lim_{x \to -3} (x^2 + x - 6)/(x^2 + 3x) = (9 - 3 - 6)/(9 - 9) = 0/0$, indeterminate.

$= \lim_{x \to -3} (x + 3)(x - 2) / [x(x + 3)] = \lim_{x \to -3} (x - 2)/x$

$= ((-3) - 2)/(-3) = 6/(-3) = \mathbf{-2}$

Line	1	2	3	4	5	Total
You						/ 5
AI						/ 5

Answer C

Plug in -3: $((-3)^2 + (-3) - 6)/((-3)^2 + 3(-3)) = (9 - 3 - 6)/(9 - 9) = 0/0$

$\lim_{x \to -3} (x + 3)(x - 2) / [(x + 3)(x + 1)] = \lim_{x \to -3} (x - 2)/(x + 1)$

$= (-3 - 2)/(-3 + 1) = -5/-2 = \mathbf{5/2}$

Line	1	2	3	4	5	Total
You						/ 5
AI						/ 5

Write your scores first. Then we'll see what the AI gives.

Questions? ddixon1@cau.edu